

What I will improve in the next one week

Author: Pankaj Yadav, Forward Deployed Software Engineer · github.com/Pankkaj64 · linkedin.com/in/pankkaj64

These are the things I want to add or fix next. I have kept them in the order I plan to do them. Each point has a **Status** line saying what is already done (8 October 2026).

1. Show failures and progress in the frontend

Status: partly done. The status strip (scanning / processing frame N with seconds / finishing / after-capture checks N of M), the header pill, the live counts and the edge-cut / move-closer guidance are in. Still open: the last frame with boxes drawn on it, and backend errors shown in red, starting with "model server not running" (a real sweep failed silently that way).

Right now if something goes wrong in the backend (model not loaded, OCR not working), the UI only shows one small line of text. It is easy to miss.

What I will do:

- Add a small status bar at the top. It will show: is the backend running, which models are loaded, which stage is running now ("reading spine 3 of 10"), how many frames are waiting.
- Show the last captured frame with the boxes drawn on it, so the user can see what was detected and what was missed while they are still standing in front of the shelf.
- If a frame fails, show the reason in red (for example "detector not installed") instead of hiding it in the guidance text.

Why: the reviewer will point the app at their own shelf. If something breaks, we should see it immediately, not after the sweep.

2. Tell the user in the frontend: this is a book / this is not a book

Status: backend done, frontend open. The blind check records what it saw and the words it read ("Blind check saw a tissue box, not a book"); the label next to each box and the move-to-items button are still to do.

Today the inventory only shows a list of books and items. The user cannot see why one box was accepted as a book and another was not.

What I will do:

- Next to every detected box, show a small label: "book", "not a book (cup)", or "not sure". This comes from the second model (Qwen) which checks the crop without knowing the label.
- If it says "not a book", show what it thinks it is, and give one button to move it to items.
- If it says "not sure", keep it in the review list with the reason written in plain words.

Why: this is the main safety check against wrong items entering the claim, and right now it is invisible to the user.

3. Replace the 2-second snapshots with proper video using MediaRecorder

Status: open. The frame budgets, the after-capture pass and cross-frame evidence pooling were done first so that video frames can be processed without stalling the sweep.

Right now the browser takes one picture every 2 seconds. Anything the camera passes between two pictures is never seen. Also, when the backend is busy with one frame, no new picture is taken at all, so the real gap can be minutes.

What I will do:

- Use the browser `MediaRecorder` API to record the whole sweep as one video (WebM or MP4) and send it to the backend in small chunks while recording.
- On the backend, read the video and take frames from it at a fixed rate (for example 2 per second), and pick the sharpest frame for each shelf.
- Keep the live preview and the voice guidance exactly as they are now.
- Keep the video file as evidence, so every frame reference can point to a time in the video.

Why: the brief says "one continuous sweep". With video, nothing is missed between snapshots and the agent can re-read a shelf from the recording without asking the user to go back.

4. Pricing in more detail

Status: partly done. Google Books works with a key, the author-name and misread checks run before pricing, and a check tool (`backend/tools/check_price_sources.py`) exists. eBay keys, percentages, availability and history are still to do.

Right now a book shows one replacement price and one used price (and most are blank until the eBay and Google Books keys are added). I want to show more about the price so the adjuster can trust it.

What I will do:

- Add eBay and Google Books keys first, so real prices come in.
- For every book show: new price, used price, and the difference in percent (for example "used is 60% below new").
- Show how many second-hand copies are available right now and the lowest / median / highest price of them, with the links.
- Show if the price is from the local market or converted from another country, and the exchange rate and date used.
- Show a small history: if the same book was priced before (another sweep or the second-country comparison), show how much the price went up or down in percent.

- Keep the rule: if there is no source, the price stays blank and goes to the review list.

Why: pricing is 20% of the score, and the reviewer wants to see where every number came from.

5. Mobile-friendly layout

Status: done. Phone, tablet, desktop and wide breakpoints, 44 px buttons, two pages (sweep and inventory), HTTPS dev server so phones on the LAN get the camera. Still to test: Chrome on Android and Safari on iPhone with the microphone permission.

The brief says the user opens the app on a phone. Today the page is designed for a laptop screen.

What I will do:

- One column layout on small screens: camera on top, the agent's message under it, then the inventory, then the transcript.
- Bigger buttons (Start, Stop, Capture now) that are easy to press with a thumb.
- Use the back camera by default on the phone (already done) and keep the screen awake during the sweep.
- Test on Chrome on Android and Safari on iPhone, including the microphone permission.

Why: a real sweep is done walking around a room with a phone, not sitting at a laptop.

Order and time

Day	Work
1	status bar, frame with boxes, errors in red
2	book / not a book labels and the move-to-items button
3–4	MediaRecorder video and frame extraction on the backend
5	price keys, price details and percentages
6	mobile layout and phone testing
7	one full sweep of my own room with all of this, fix what breaks